AFM - Atomic Force Microscope

Signature Sheet

Student's Name _____ Partner's Name _____

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. How does an AFM differ from other microscopy techniques, e.g. optical, SEM, and STM? What are the advantages of the AFM? (You can check how many variants the AFM has in the Scanning Probe Microscopy Wikipedia page and think about why.)

2. How does the AFM read the feedback signal — where does the feedback come from? We have a Vibrating Mode and a Non-Vibrating Mode: what quantity feeds the feedback loop in each?

3. Draw, and briefly explain the different features of the typical Force-Distance curve for a material. Be sure to note the attractive and repulsive regions. Based on this curve, how does the deflection signal behave in contact (Non-Vibrating) mode as the tip approaches the sample? Note the difference in *sign* between the attractive and the repulsive region.

4. In Vibrating (amplitude-modulation, or "tapping") mode, the cantilever is driven at a fixed DRIVING FREQUENCY near its resonance, and the Z feedback holds the oscillation amplitude at the SET-POINT. Explain what these two parameters mean, and how the oscillation amplitude changes as the tip approaches the surface. In practice, to image in the *repulsive* regime, the driving frequency is usually set slightly *below* the resonance peak (on its left shoulder), with a setpoint slightly smaller than the free-oscillation amplitude — why? (Hint: model the tip–sample interaction as a small change of the cantilever's spring constant. Using the force–distance curve of the previous question, decide which way the resonance peak shifts in the attractive and in the repulsive region. And one more thing to think about: in Non-Vibrating mode the SETPOINT is an absolute value, but in Vibrating mode it is a *percentage* — a percentage of what, and why a percentage?)

5. The AFM belongs to the family of scanning probe microscopy, in which a sharp probe is positioned by piezoelectric elements. What is a piezoelectric scanner, and why are piezos used for this job? Which motions in our AFM need a piezo, and what does each of them do?

6. Explain what the important parameters do in the Pre-Scan, Topo Scan, and System tab (frequency select, manual Z motor control, automated tip approach, range check, scan lines, scan size, rotation, Z feedback, display, HV Z gain, XY parameters, tip approach parameters, calibration)

7. Give examples of two scenarios where you are likely to break a tip.

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

On day 2 of this lab, you should have the ability to scan samples accurately and reliably. Show a processed topography scan of the calibration sample to an instructor. Take a profile of a row of features to show that your measurements and calibrations are correct. How close are your measurements compared with the datasheet?

1. What is a scanned image's orientation relative to the camera's view orientation? How does Scan Rotation change this?
2. Which direction does the scanned image move if you move the physical sample to the left? What about if you move the physical sample up?

Staff Signature _____ Date _____

Completed by day 2 of lab? (Circle one) Yes / No

Checkpoint Signatures

1. Alignments

   Staff Signature _____

2. Pre-Scan Settings

   Staff Signature _____

3. Calibration

   Staff Signature _____

4. CD/DVD

Staff Signature _____

5. Boltzmann's Constant

Staff Signature _____

Station Clean-Up Check-Off

Before leaving, ensure your station is clean and ready for the next group. The GSI must verify and sign off on the following:

1. Any BNC cables that you have been moving are unplugged; return things to the state that you found them.
2. AFM prepared samples are removed and the sample disks are returned to the bag. CD/DVD/Blu-Ray scraps are thrown out if small, put in the bin if large.
3. All used pipette tips go into trash and glass sample slides go into glass disposal container; pipettes are put back to the shelf.
4. Workspace cleared of personal belongings, papers, and debris.
5. Any spills or hazards cleaned up; no left-over onion in the fridge or on the bench.
6. Microscope is clear of mounted sample, turned off and covered by the plastic bag properly if applicable.
7. Restart the computer at your station so that any software updates are applied.
8. Station is clean and ready for the next group. Staff discretion for additional cleanup that is necessary applies.

Staff Signature _____ Date _____